

10(a)	eight from: 1 ADH reaches (cells of) collecting duct ; 2 binds to receptor on cell surface membrane ; 3 enzyme-controlled reactions / enzyme cascade ; 4 (active) phosphorylase ; 5 (which causes) vesicles with, aquaporins / water channels ; 6 to, move to / fuse with, (cell surface) membrane ; 7 increase (membrane) permeability (to water) ; 8 (more) water moves out of collecting duct ; 9 by osmosis / down water potential gradient ; 10 (then) into blood ; 11 urine (more) concentrated / small volume of urine ; 12 AVP ; e.g. role of loop of Henle in creating water potential gradient movement of urea increases water potential gradient	8
10(b)	<i>seven from:</i> 1 (glucagon) acts as cell signalling molecule ; 2 binds to receptor on cell surface membrane ; 3 liver cell ; 4 G-protein activated ; 5 adenylyl cyclase activated ; 6 formation of cAMP ; 7 (which is) second messenger ; 8 enzyme-controlled reactions / enzyme cascade ; 9 signal amplified ; 10 glycogen hydrolysed to glucose / glycogenolysis ; 11 (glucose released) into blood ; 12 AVP ; e.g. ref. to glucose exits via GLUT2 proteins / gluconeogenesis	7